

1) Для заданной на схеме schema-lab2 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета

```
vIOS-L2-01>enable
```

```
vIOS-L2-01#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
vIOS-L2-01(config)#spanning-tree vlan 1 priority 0
```

```
vIOS-L2-01(config)#exit
```

```
vIOS-L2-01#
```

```
*May 10 13:36:23.377: %SYS-5-CONFIG_I: Configured from console by console
```

2) Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать

```
PC1> ip 192.168.0.1/24
```

```
Checking for duplicate address...
```

```
PC1 : 192.168.0.1 255.255.255.0
```

```
PC2> ip 192.168.0.2/24
```

```
Checking for duplicate address...
```

```
PC2 : 192.168.0.2 255.255.255.0
```

```
PC3> ip 192.168.0.3/24
```

```
Checking for duplicate address...
```

```
PC3 : 192.168.0.3 255.255.255.0
```

```
PC4> ip 192.168.0.4/24
```

```
Checking for duplicate address...
```

```
PC4 : 192.168.0.4 255.255.255.0
```

```
PC5> ip 192.168.0.5/24
```

Checking for duplicate address...

PC5 : 192.168.0.5 255.255.255.0

PC6> ip 192.168.0.6/24

Checking for duplicate address...

PC6 : 192.168.0.6 255.255.255.0

От PC1:

PC1> ping 192.168.0.2

84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=0.900 ms

84 bytes from 192.168.0.2 icmp_seq=2 ttl=64 time=5.543 ms

84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=0.654 ms

84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=1.058 ms

84 bytes from 192.168.0.2 icmp_seq=5 ttl=64 time=0.705 ms

PC1> ping 192.168.0.3

84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=16.547 ms

84 bytes from 192.168.0.3 icmp_seq=2 ttl=64 time=4.905 ms

84 bytes from 192.168.0.3 icmp_seq=3 ttl=64 time=10.323 ms

84 bytes from 192.168.0.3 icmp_seq=4 ttl=64 time=7.151 ms

84 bytes from 192.168.0.3 icmp_seq=5 ttl=64 time=5.953 ms

PC1> ping 192.168.0.4

84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=13.307 ms

84 bytes from 192.168.0.4 icmp_seq=2 ttl=64 time=10.827 ms

84 bytes from 192.168.0.4 icmp_seq=3 ttl=64 time=1.974 ms
84 bytes from 192.168.0.4 icmp_seq=4 ttl=64 time=11.265 ms
84 bytes from 192.168.0.4 icmp_seq=5 ttl=64 time=10.324 ms

PC1> ping 192.168.0.5

84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=11.281 ms
84 bytes from 192.168.0.5 icmp_seq=2 ttl=64 time=2.443 ms
84 bytes from 192.168.0.5 icmp_seq=3 ttl=64 time=10.996 ms
84 bytes from 192.168.0.5 icmp_seq=4 ttl=64 time=7.169 ms
84 bytes from 192.168.0.5 icmp_seq=5 ttl=64 time=9.861 ms

PC1> ping 192.168.0.6

84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=6.070 ms
84 bytes from 192.168.0.6 icmp_seq=2 ttl=64 time=10.553 ms
84 bytes from 192.168.0.6 icmp_seq=3 ttl=64 time=2.295 ms
84 bytes from 192.168.0.6 icmp_seq=4 ttl=64 time=11.358 ms
84 bytes from 192.168.0.6 icmp_seq=5 ttl=64 time=13.409 ms

От PC2:

PC2> ping 192.168.0.1

84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=0.792 ms
84 bytes from 192.168.0.1 icmp_seq=2 ttl=64 time=1.675 ms
84 bytes from 192.168.0.1 icmp_seq=3 ttl=64 time=7.257 ms
84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=7.741 ms
84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=0.679 ms

PC2> ping 192.168.0.3

84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=5.359 ms

84 bytes from 192.168.0.3 icmp_seq=2 ttl=64 time=9.831 ms

84 bytes from 192.168.0.3 icmp_seq=3 ttl=64 time=15.694 ms

84 bytes from 192.168.0.3 icmp_seq=4 ttl=64 time=3.965 ms

84 bytes from 192.168.0.3 icmp_seq=5 ttl=64 time=4.834 ms

PC2> ping 192.168.0.4

84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=16.173 ms

84 bytes from 192.168.0.4 icmp_seq=2 ttl=64 time=3.542 ms

84 bytes from 192.168.0.4 icmp_seq=3 ttl=64 time=4.128 ms

84 bytes from 192.168.0.4 icmp_seq=4 ttl=64 time=13.569 ms

84 bytes from 192.168.0.4 icmp_seq=5 ttl=64 time=1.853 ms

PC2> ping 192.168.0.5

84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=14.577 ms

84 bytes from 192.168.0.5 icmp_seq=2 ttl=64 time=10.324 ms

84 bytes from 192.168.0.5 icmp_seq=3 ttl=64 time=9.552 ms

84 bytes from 192.168.0.5 icmp_seq=4 ttl=64 time=4.793 ms

84 bytes from 192.168.0.5 icmp_seq=5 ttl=64 time=4.235 ms

PC2> ping 192.168.0.6

84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=8.087 ms

84 bytes from 192.168.0.6 icmp_seq=2 ttl=64 time=4.075 ms

84 bytes from 192.168.0.6 icmp_seq=3 ttl=64 time=3.730 ms
84 bytes from 192.168.0.6 icmp_seq=4 ttl=64 time=4.229 ms
84 bytes from 192.168.0.6 icmp_seq=5 ttl=64 time=4.120 ms

От PC3:

PC3> ping 192.168.0.1

84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=11.597 ms
84 bytes from 192.168.0.1 icmp_seq=2 ttl=64 time=4.650 ms
84 bytes from 192.168.0.1 icmp_seq=3 ttl=64 time=5.553 ms
84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=6.698 ms
84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=3.557 ms

PC3> ping 192.168.0.2

84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=4.409 ms
84 bytes from 192.168.0.2 icmp_seq=2 ttl=64 time=9.441 ms
84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=10.096 ms
84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=17.002 ms
84 bytes from 192.168.0.2 icmp_seq=5 ttl=64 time=8.073 ms

PC3> ping 192.168.0.4

84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=2.527 ms
84 bytes from 192.168.0.4 icmp_seq=2 ttl=64 time=5.212 ms
84 bytes from 192.168.0.4 icmp_seq=3 ttl=64 time=1.096 ms
84 bytes from 192.168.0.4 icmp_seq=4 ttl=64 time=4.212 ms
84 bytes from 192.168.0.4 icmp_seq=5 ttl=64 time=1.866 ms

PC3> ping 192.168.0.5

84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=9.554 ms

84 bytes from 192.168.0.5 icmp_seq=2 ttl=64 time=6.155 ms

84 bytes from 192.168.0.5 icmp_seq=3 ttl=64 time=2.715 ms

84 bytes from 192.168.0.5 icmp_seq=4 ttl=64 time=2.758 ms

84 bytes from 192.168.0.5 icmp_seq=5 ttl=64 time=3.169 ms

PC3> ping 192.168.0.6

84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=13.760 ms

84 bytes from 192.168.0.6 icmp_seq=2 ttl=64 time=8.802 ms

84 bytes from 192.168.0.6 icmp_seq=3 ttl=64 time=5.749 ms

84 bytes from 192.168.0.6 icmp_seq=4 ttl=64 time=9.870 ms

84 bytes from 192.168.0.6 icmp_seq=5 ttl=64 time=5.437 ms

От PC4:

PC4> ping 192.168.0.1

84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=8.699 ms

84 bytes from 192.168.0.1 icmp_seq=2 ttl=64 time=4.401 ms

84 bytes from 192.168.0.1 icmp_seq=3 ttl=64 time=9.050 ms

84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=10.332 ms

84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=19.773 ms

PC4> ping 192.168.0.2

84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=16.933 ms
84 bytes from 192.168.0.2 icmp_seq=2 ttl=64 time=5.690 ms
84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=5.708 ms
84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=6.232 ms
84 bytes from 192.168.0.2 icmp_seq=5 ttl=64 time=10.778 ms

PC4> ping 192.168.0.3

84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=0.796 ms
84 bytes from 192.168.0.3 icmp_seq=2 ttl=64 time=3.492 ms
84 bytes from 192.168.0.3 icmp_seq=3 ttl=64 time=9.238 ms
84 bytes from 192.168.0.3 icmp_seq=4 ttl=64 time=7.405 ms
84 bytes from 192.168.0.3 icmp_seq=5 ttl=64 time=1.731 ms

PC4> ping 192.168.0.5

84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=10.648 ms
84 bytes from 192.168.0.5 icmp_seq=2 ttl=64 time=3.799 ms
84 bytes from 192.168.0.5 icmp_seq=3 ttl=64 time=9.162 ms
84 bytes from 192.168.0.5 icmp_seq=4 ttl=64 time=16.081 ms
84 bytes from 192.168.0.5 icmp_seq=5 ttl=64 time=6.794 ms

PC4> ping 192.168.0.6

84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=9.781 ms
84 bytes from 192.168.0.6 icmp_seq=2 ttl=64 time=10.722 ms
84 bytes from 192.168.0.6 icmp_seq=3 ttl=64 time=4.900 ms
84 bytes from 192.168.0.6 icmp_seq=4 ttl=64 time=2.705 ms
84 bytes from 192.168.0.6 icmp_seq=5 ttl=64 time=3.172 ms

От PC5:

PC5> ping 192.168.0.1

84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=9.200 ms

84 bytes from 192.168.0.1 icmp_seq=2 ttl=64 time=4.722 ms

84 bytes from 192.168.0.1 icmp_seq=3 ttl=64 time=8.293 ms

84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=7.869 ms

84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=7.930 ms

PC5> ping 192.168.0.2

84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=20.840 ms

84 bytes from 192.168.0.2 icmp_seq=2 ttl=64 time=9.468 ms

84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=8.591 ms

84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=7.158 ms

84 bytes from 192.168.0.2 icmp_seq=5 ttl=64 time=8.203 ms

PC5> ping 192.168.0.3

84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=7.377 ms

84 bytes from 192.168.0.3 icmp_seq=2 ttl=64 time=6.833 ms

84 bytes from 192.168.0.3 icmp_seq=3 ttl=64 time=9.146 ms

84 bytes from 192.168.0.3 icmp_seq=4 ttl=64 time=7.014 ms

84 bytes from 192.168.0.3 icmp_seq=5 ttl=64 time=9.332 ms

PC5> ping 192.168.0.4

84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=12.624 ms
84 bytes from 192.168.0.4 icmp_seq=2 ttl=64 time=8.297 ms
84 bytes from 192.168.0.4 icmp_seq=3 ttl=64 time=5.330 ms
84 bytes from 192.168.0.4 icmp_seq=4 ttl=64 time=7.749 ms
84 bytes from 192.168.0.4 icmp_seq=5 ttl=64 time=6.879 ms

PC5> ping 192.168.0.6

84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=2.680 ms
84 bytes from 192.168.0.6 icmp_seq=2 ttl=64 time=0.989 ms
84 bytes from 192.168.0.6 icmp_seq=3 ttl=64 time=5.777 ms
84 bytes from 192.168.0.6 icmp_seq=4 ttl=64 time=2.227 ms
84 bytes from 192.168.0.6 icmp_seq=5 ttl=64 time=1.072 ms

Or PC6:

PC6> ping 192.168.0.1

84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=9.332 ms
84 bytes from 192.168.0.1 icmp_seq=2 ttl=64 time=16.986 ms
84 bytes from 192.168.0.1 icmp_seq=3 ttl=64 time=7.440 ms
84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=6.926 ms
84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=8.692 ms

PC6> ping 192.168.0.2

84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=6.982 ms
84 bytes from 192.168.0.2 icmp_seq=2 ttl=64 time=4.930 ms
84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=7.745 ms

84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=13.150 ms

84 bytes from 192.168.0.2 icmp_seq=5 ttl=64 time=4.681 ms

PC6> ping 192.168.0.3

84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=11.897 ms

84 bytes from 192.168.0.3 icmp_seq=2 ttl=64 time=5.252 ms

84 bytes from 192.168.0.3 icmp_seq=3 ttl=64 time=8.256 ms

84 bytes from 192.168.0.3 icmp_seq=4 ttl=64 time=7.133 ms

84 bytes from 192.168.0.3 icmp_seq=5 ttl=64 time=9.224 ms

PC6> ping 192.168.0.4

84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=8.057 ms

84 bytes from 192.168.0.4 icmp_seq=2 ttl=64 time=11.470 ms

84 bytes from 192.168.0.4 icmp_seq=3 ttl=64 time=12.529 ms

84 bytes from 192.168.0.4 icmp_seq=4 ttl=64 time=8.741 ms

84 bytes from 192.168.0.4 icmp_seq=5 ttl=64 time=9.463 ms

PC6> ping 192.168.0.5

84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=2.756 ms

84 bytes from 192.168.0.5 icmp_seq=2 ttl=64 time=9.259 ms

84 bytes from 192.168.0.5 icmp_seq=3 ttl=64 time=9.394 ms

84 bytes from 192.168.0.5 icmp_seq=4 ttl=64 time=1.801 ms

84 bytes from 192.168.0.5 icmp_seq=5 ttl=64 time=6.355 ms

3) На изображении схемы отметить VID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл

Коммутатор 1:

vIOS-L2-01>show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 1

Address 0c68.b492.0000

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 1 (priority 0 sys-id-ext 1)

Address 0c68.b492.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type

Gi0/0	Desg	FWD	4	128.1	Shr
Gi0/1	Desg	FWD	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr
Gi1/2	Desg	FWD	4	128.7	Shr
Gi1/3	Desg	FWD	4	128.8	Shr

Коммутатор 2:

vIOS-L2-01>show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 1

Address 0c68.b492.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0ce1.76e6.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type

Gi0/0	Root	FWD	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr
Gi0/2	Altn	BLK	4	128.3	Shr
Gi0/3	Altn	BLK	4	128.4	Shr
Gi1/0	Altn	BLK	4	128.5	Shr
Gi1/1	Altn	BLK	4	128.6	Shr
Gi1/2	Altn	BLK	4	128.7	Shr
Gi1/3	Altn	BLK	4	128.8	Shr
Gi2/0	Desg	FWD	4	128.9	Shr

Коммутатор 3:

vIOS-L2-01>show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 1

Address 0c68.b492.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c91.9c21.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Root	FWD	4	128.1	Shr	
-------	------	-----	---	-------	-----	--

Gi0/1	Altn	BLK	4	128.2	Shr	
-------	------	-----	---	-------	-----	--

Gi0/2	Desg	FWD	4	128.3	Shr	
-------	------	-----	---	-------	-----	--

Gi0/3	Desg	FWD	4	128.4	Shr	
-------	------	-----	---	-------	-----	--

Gi1/0	Desg	FWD	4	128.5	Shr	
-------	------	-----	---	-------	-----	--

Gi1/1	Desg	FWD	4	128.6	Shr	
-------	------	-----	---	-------	-----	--

Коммутатор 4:

vIOS-L2-01>show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 1

Address 0c68.b492.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0cb2.1eb9.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Root	FWD	4	128.1		Shr
-------	------	-----	---	-------	--	-----

Gi0/1	Altn	BLK	4	128.2		Shr
-------	------	-----	---	-------	--	-----

Gi0/2	Desg	FWD	4	128.3		Shr
-------	------	-----	---	-------	--	-----

Gi0/3	Desg	FWD	4	128.4		Shr
-------	------	-----	---	-------	--	-----

Gi1/0	Desg	FWD	4	128.5		Shr
-------	------	-----	---	-------	--	-----

Gi1/1	Desg	FWD	4	128.6		Shr
-------	------	-----	---	-------	--	-----

Коммутатор 5:

vIOS-L2-01>show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 1

Address 0c68.b492.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

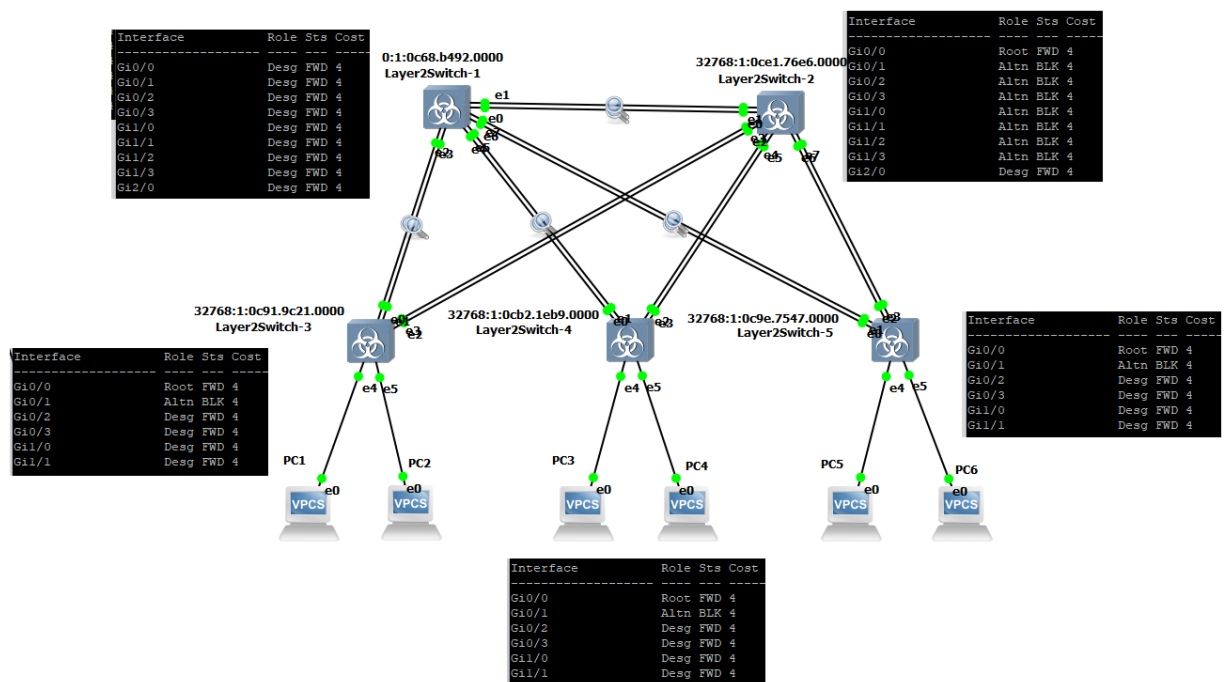
Address 0c9e.7547.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Root	FWD	4	128.1	Shr	
Gi0/1	Altn	BLK	4	128.2	Shr	
Gi0/2	Desg	FWD	4	128.3	Shr	
Gi0/3	Desg	FWD	4	128.4	Shr	
Gi1/0	Desg	FWD	4	128.5	Shr	
Gi1/1	Desg	FWD	4	128.6	Shr	



4) При помощи wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет

К коммутатору 2:

К коммутатору 5:

e6:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:68:b4:92:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:68:b4:92:00:00 Cost = 0 Port = 0x8007
2	0.008889	0c:68:b4:92:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:68:b4:92:00:00 Cost = 0 Port = 0x8007
3	0.013921	0c:68:b4:92:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:68:b4:92:00:00 Cost = 0 Port = 0x8007
4	0.042645	0c:68:b4:92:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8007
5	0.992068	0c:9e:75:47:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:9e:75:47:00:00 Cost = 0 Port = 0x8001
6	0.992315	0c:9e:75:47:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:9e:75:47:00:00 Cost = 0 Port = 0x8001
7	0.994029	0c:9e:75:47:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:9e:75:47:00:00 Cost = 0 Port = 0x8001
8	1.043486	0c:68:b4:92:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8007
9	2.002642	0c:68:b4:92:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:68:b4:92:00:00 Cost = 0 Port = 0x8007
10	2.011577	0c:68:b4:92:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:68:b4:92:00:00 Cost = 0 Port = 0x8007
11	2.016683	0c:68:b4:92:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:68:b4:92:00:00 Cost = 0 Port = 0x8007
12	2.045357	0c:68:b4:92:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8007
13	2.995065	0c:9e:75:47:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:9e:75:47:00:00 Cost = 0 Port = 0x8001
14	2.995247	0c:9e:75:47:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:9e:75:47:00:00 Cost = 0 Port = 0x8001
15	2.996720	0c:9e:75:47:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:9e:75:47:00:00 Cost = 0 Port = 0x8001
16	3.045188	0c:68:b4:92:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8007
17	4.006451	0c:68:b4:92:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:68:b4:92:00:00 Cost = 0 Port = 0x8007
18	4.015305	0c:68:b4:92:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:68:b4:92:00:00 Cost = 0 Port = 0x8007
19	4.020243	0c:68:b4:92:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:68:b4:92:00:00 Cost = 0 Port = 0x8007
20	4.049070	0c:68:b4:92:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8007
21	4.998450	0c:9e:75:47:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:9e:75:47:00:00 Cost = 0 Port = 0x8001
22	4.998881	0c:9e:75:47:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:9e:75:47:00:00 Cost = 0 Port = 0x8001
23	5.000402	0c:9e:75:47:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:9e:75:47:00:00 Cost = 0 Port = 0x8001
24	5.051888	0c:68:b4:92:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8007

e7:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
2	0.007033	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
3	0.013064	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
4	0.039715	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
5	0.940113	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002
6	0.941354	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002
7	0.942314	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002
8	1.040544	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
9	2.003660	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
10	2.010754	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
11	2.016631	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
12	2.043409	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
14	2.941834	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002
15	2.943031	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002
16	2.944032	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002
17	3.043269	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
19	4.006337	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
20	4.013354	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
21	4.019308	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
22	4.046112	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
24	4.942528	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002
25	4.944027	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002
26	4.945185	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002
27	5.045988	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008

Для настроенного VLAN 1 передаётся как корень первый коммутатор с приоритетом 0. Другие коммутаторы не создают Hello-пакетов для VLAN 1, а только ретранслируют пакеты от первого коммутатора другим, поэтому на линках с первым коммутатором нет входящих Hello-пакетов для VLAN 1.

5) Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл

Коммутатор 5:

```
vIOS-L2-01>enable
```

```
vIOS-L2-01#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
vIOS-L2-01(config)#int gi0/0
```

```
vIOS-L2-01(config-if)#spanning-tree vlan 1 cost 5
```

```
vIOS-L2-01(config-if)#exit
```

```
vIOS-L2-01(config)#exit
```

```
vIOS-L2-01#
```

```
*May 10 15:57:46.860: %SYS-5-CONFIG_I: Configured from console by console
```

```
vIOS-L2-01#show spanning-tree vlan 1
```

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 1

 Address 0c68.b492.0000

 Cost 4

 Port 2 (GigabitEthernet0/1)

 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

 Address 0c9e.7547.0000

 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

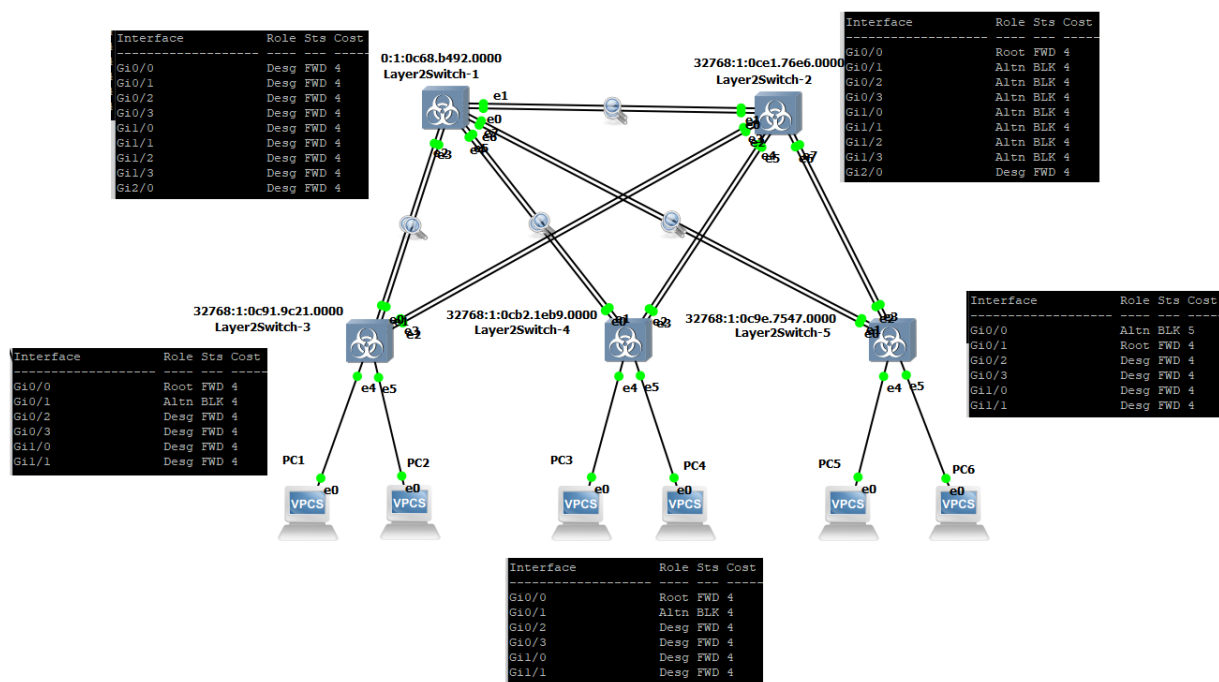
 Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Altn	BLK	5	128.1	Shr	
-------	------	-----	---	-------	-----	--

Gi0/1	Root	FWD	4	128.2	Shr	
-------	------	-----	---	-------	-----	--

Gi0/2	Desg FWD 4	128.3	Shr
Gi0/3	Desg FWD 4	128.4	Shr
Gi1/0	Desg FWD 4	128.5	Shr
Gi1/1	Desg FWD 4	128.6	Shr



6) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств

7*) Опциональное задание: заменить STP на RSTP (IEEE 802.1w), повторить 1-6, отметить резервные порты в п.3 и п.5, отличие работы протокола RSTP от протокола STP в п.4

7-1) Для заданной на схеме schema-lab2 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета

На всех коммутаторах:

vIOS-L2-01>enable

vIOS-L2-01#conf t

Enter configuration commands, one per line. End with CNTL/Z.

```
vIOS-L2-01(config)#spanning-tree mode rapid-pvst
```

```
vIOS-L2-01(config)#exit
```

```
vIOS-L2-01#
```

*May 10 16:59:05.876: %SYS-5-CONFIG_I: Configured from console by console

На коммутаторах 3-5:

```
vIOS-L2-01#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
vIOS-L2-01(config)#int gi1/0
```

```
vIOS-L2-01(config-if)#spanning-tree portfast
```

%Warning: portfast should only be enabled on ports connected to a single host. Connecting hubs, concentrators, switches, bridges, etc... to this interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet1/0 but will only have effect when the interface is in a non-trunking mode.

```
vIOS-L2-01(config-if)#exit
```

```
vIOS-L2-01(config)#int gi1/1
```

```
vIOS-L2-01(config-if)#spanning-tree portfast
```

%Warning: portfast should only be enabled on ports connected to a single host. Connecting hubs, concentrators, switches, bridges, etc... to this interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet1/1 but will only have effect when the interface is in a non-trunking mode.

```
vIOS-L2-01(config-if)#exit
```

```
vIOS-L2-01(config)#exit
```

7-2) Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать

От PC1:

```
PC1> ping 192.168.0.2
```

```
84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=7.454 ms
```

```
84 bytes from 192.168.0.2 icmp_seq=2 ttl=64 time=7.004 ms
```

```
84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=3.342 ms
```

```
84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=0.715 ms
```

```
84 bytes from 192.168.0.2 icmp_seq=5 ttl=64 time=1.435 ms
```

```
PC1> ping 192.168.0.3
```

```
84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=13.944 ms
```

```
192.168.0.3 icmp_seq=2 timeout
```

```
192.168.0.3 icmp_seq=3 timeout
```

```
192.168.0.3 icmp_seq=4 timeout
```

```
192.168.0.3 icmp_seq=5 timeout
```

```
PC1> ping 192.168.0.4
```

```
84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=2.602 ms
```

```
192.168.0.4 icmp_seq=2 timeout
```

```
192.168.0.4 icmp_seq=3 timeout
```

192.168.0.4 icmp_seq=4 timeout

192.168.0.4 icmp_seq=5 timeout

PC1> ping 192.168.0.5

84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=4.577 ms

84 bytes from 192.168.0.5 icmp_seq=2 ttl=64 time=9.273 ms

84 bytes from 192.168.0.5 icmp_seq=3 ttl=64 time=4.259 ms

192.168.0.5 icmp_seq=4 timeout

192.168.0.5 icmp_seq=5 timeout

PC1> ping 192.168.0.6

84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=16.725 ms

192.168.0.6 icmp_seq=2 timeout

192.168.0.6 icmp_seq=3 timeout

192.168.0.6 icmp_seq=4 timeout

192.168.0.6 icmp_seq=5 timeout

От PC2:

PC2> ping 192.168.0.1

84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=4.218 ms

84 bytes from 192.168.0.1 icmp_seq=2 ttl=64 time=2.802 ms

84 bytes from 192.168.0.1 icmp_seq=3 ttl=64 time=7.474 ms

84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=3.724 ms

84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=2.280 ms

PC2> ping 192.168.0.3

84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=4.405 ms

192.168.0.3 icmp_seq=2 timeout

192.168.0.3 icmp_seq=3 timeout

192.168.0.3 icmp_seq=4 timeout

192.168.0.3 icmp_seq=5 timeout

PC2> ping 192.168.0.4

84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=7.957 ms

192.168.0.4 icmp_seq=2 timeout

192.168.0.4 icmp_seq=3 timeout

192.168.0.4 icmp_seq=4 timeout

192.168.0.4 icmp_seq=5 timeout

PC2> ping 192.168.0.5

84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=14.306 ms

84 bytes from 192.168.0.5 icmp_seq=2 ttl=64 time=7.685 ms

192.168.0.5 icmp_seq=3 timeout

192.168.0.5 icmp_seq=4 timeout

192.168.0.5 icmp_seq=5 timeout

PC2> ping 192.168.0.6

84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=8.567 ms

192.168.0.6 icmp_seq=2 timeout

192.168.0.6 icmp_seq=3 timeout

192.168.0.6 icmp_seq=4 timeout

192.168.0.6 icmp_seq=5 timeout

От PC3:

PC3> ping 192.168.0.1

192.168.0.1 icmp_seq=1 timeout

192.168.0.1 icmp_seq=2 timeout

192.168.0.1 icmp_seq=3 timeout

84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=9.567 ms

84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=5.868 ms

PC3> ping 192.168.0.2

84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=14.368 ms

192.168.0.2 icmp_seq=2 timeout

192.168.0.2 icmp_seq=3 timeout

192.168.0.2 icmp_seq=4 timeout

192.168.0.2 icmp_seq=5 timeout

PC3> ping 192.168.0.4

84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=8.520 ms

84 bytes from 192.168.0.4 icmp_seq=2 ttl=64 time=2.767 ms

84 bytes from 192.168.0.4 icmp_seq=3 ttl=64 time=4.342 ms

84 bytes from 192.168.0.4 icmp_seq=4 ttl=64 time=7.885 ms

84 bytes from 192.168.0.4 icmp_seq=5 ttl=64 time=7.018 ms

PC3> ping 192.168.0.5

84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=3.967 ms

192.168.0.5 icmp_seq=2 timeout

192.168.0.5 icmp_seq=3 timeout

192.168.0.5 icmp_seq=4 timeout

192.168.0.5 icmp_seq=5 timeout

PC3> ping 192.168.0.6

84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=6.537 ms

192.168.0.6 icmp_seq=2 timeout

192.168.0.6 icmp_seq=3 timeout

192.168.0.6 icmp_seq=4 timeout

192.168.0.6 icmp_seq=5 timeout

От PC4:

PC4> ping 192.168.0.1

84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=7.140 ms

192.168.0.1 icmp_seq=2 timeout

192.168.0.1 icmp_seq=3 timeout

192.168.0.1 icmp_seq=4 timeout

192.168.0.1 icmp_seq=5 timeout

PC4> ping 192.168.0.2

84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=13.868 ms

192.168.0.2 icmp_seq=2 timeout

192.168.0.2 icmp_seq=3 timeout

192.168.0.2 icmp_seq=4 timeout

84 bytes from 192.168.0.2 icmp_seq=5 ttl=64 time=9.619 ms

PC4> ping 192.168.0.3

84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=5.522 ms

84 bytes from 192.168.0.3 icmp_seq=2 ttl=64 time=8.883 ms

84 bytes from 192.168.0.3 icmp_seq=3 ttl=64 time=7.298 ms

84 bytes from 192.168.0.3 icmp_seq=4 ttl=64 time=7.734 ms

84 bytes from 192.168.0.3 icmp_seq=5 ttl=64 time=4.838 ms

PC4> ping 192.168.0.5

84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=7.782 ms

192.168.0.5 icmp_seq=2 timeout

192.168.0.5 icmp_seq=3 timeout

192.168.0.5 icmp_seq=4 timeout

192.168.0.5 icmp_seq=5 timeout

PC4> ping 192.168.0.6

84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=11.074 ms

192.168.0.6 icmp_seq=2 timeout

192.168.0.6 icmp_seq=3 timeout

84 bytes from 192.168.0.6 icmp_seq=4 ttl=64 time=10.870 ms

192.168.0.6 icmp_seq=5 timeout

От PC5:

PC5> ping 192.168.0.1

84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=9.210 ms

192.168.0.1 icmp_seq=2 timeout

84 bytes from 192.168.0.1 icmp_seq=3 ttl=64 time=18.356 ms

84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=11.114 ms

84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=7.482 ms

PC5> ping 192.168.0.2

84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=6.210 ms

84 bytes from 192.168.0.2 icmp_seq=2 ttl=64 time=16.785 ms

84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=3.502 ms

84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=3.288 ms

192.168.0.2 icmp_seq=5 timeout

PC5> ping 192.168.0.3

84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=11.700 ms

192.168.0.3 icmp_seq=2 timeout

192.168.0.3 icmp_seq=3 timeout

192.168.0.3 icmp_seq=4 timeout

192.168.0.3 icmp_seq=5 timeout

PC5> ping 192.168.0.4

84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=5.114 ms

192.168.0.4 icmp_seq=2 timeout

192.168.0.4 icmp_seq=3 timeout

192.168.0.4 icmp_seq=4 timeout

192.168.0.4 icmp_seq=5 timeout

PC5> ping 192.168.0.6

84 bytes from 192.168.0.6 icmp_seq=1 ttl=64 time=0.649 ms

84 bytes from 192.168.0.6 icmp_seq=2 ttl=64 time=3.836 ms

84 bytes from 192.168.0.6 icmp_seq=3 ttl=64 time=1.537 ms

84 bytes from 192.168.0.6 icmp_seq=4 ttl=64 time=3.121 ms

84 bytes from 192.168.0.6 icmp_seq=5 ttl=64 time=0.615 ms

От PC6:

PC6> ping 192.168.0.1

84 bytes from 192.168.0.1 icmp_seq=1 ttl=64 time=6.266 ms

84 bytes from 192.168.0.1 icmp_seq=2 ttl=64 time=7.705 ms

192.168.0.1 icmp_seq=3 timeout

84 bytes from 192.168.0.1 icmp_seq=4 ttl=64 time=7.730 ms

84 bytes from 192.168.0.1 icmp_seq=5 ttl=64 time=8.378 ms

PC6> ping 192.168.0.2

84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=19.011 ms

192.168.0.2 icmp_seq=2 timeout

84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=8.238 ms

84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=7.820 ms

192.168.0.2 icmp_seq=5 timeout

PC6> ping 192.168.0.3

84 bytes from 192.168.0.3 icmp_seq=1 ttl=64 time=6.877 ms

192.168.0.3 icmp_seq=2 timeout

192.168.0.3 icmp_seq=3 timeout

192.168.0.3 icmp_seq=4 timeout

192.168.0.3 icmp_seq=5 timeout

PC6> ping 192.168.0.4

84 bytes from 192.168.0.4 icmp_seq=1 ttl=64 time=13.187 ms

192.168.0.4 icmp_seq=2 timeout

192.168.0.4 icmp_seq=3 timeout

192.168.0.4 icmp_seq=4 timeout

84 bytes from 192.168.0.4 icmp_seq=5 ttl=64 time=6.148 ms

PC6> ping 192.168.0.5

84 bytes from 192.168.0.5 icmp_seq=1 ttl=64 time=7.384 ms

84 bytes from 192.168.0.5 icmp_seq=2 ttl=64 time=6.904 ms

84 bytes from 192.168.0.5 icmp_seq=3 ttl=64 time=7.047 ms

84 bytes from 192.168.0.5 icmp_seq=4 ttl=64 time=4.560 ms

84 bytes from 192.168.0.5 icmp_seq=5 ttl=64 time=1.809 ms

7-3) На изображении схемы отметить VID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл

Коммутатор 1:

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 1

Address 0c68.b492.0000

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 1 (priority 0 sys-id-ext 1)

Address 0c68.b492.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type

Gi0/0	Desg	FWD	4	128.1	Shr
Gi0/1	Desg	FWD	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr
Gi1/2	Desg	FWD	4	128.7	Shr
Gi1/3	Desg	FWD	4	128.8	Shr
Gi2/0	Desg	FWD	4	128.9	Shr

Коммутатор 2:

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 1

Address 0c68.b492.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0ce1.76e6.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type

Gi0/0	Root	FWD	4	128.1	Shr	
Gi0/1	Altn	BLK	4	128.2	Shr	
Gi0/2	Altn	BLK	4	128.3	Shr	
Gi0/3	Altn	BLK	4	128.4	Shr	
Gi1/0	Altn	BLK	4	128.5	Shr	
Gi1/1	Altn	BLK	4	128.6	Shr	
Gi1/2	Altn	BLK	4	128.7	Shr	
Gi1/3	Altn	BLK	4	128.8	Shr	
Gi2/0	Desg	BLK	4	128.9	Shr	

Коммутатор 3:

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 1

Address 0c68.b492.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c91.9c21.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type

Gi0/0	Root	FWD	4	128.1	Shr	
Gi0/1	Altn	BLK	4	128.2	Shr	
Gi0/2	Desg	BLK	4	128.3	Shr	
Gi0/3	Desg	BLK	4	128.4	Shr	
Gi1/0	Desg	FWD	4	128.5	Shr	Edge
Gi1/1	Desg	FWD	4	128.6	Shr	Edge

Коммутатор 4:

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 1

Address 0c68.b492.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0cb2.1eb9.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type

Gi0/0	Root	FWD	4	128.1	Shr	
Gi0/1	Altn	BLK	4	128.2	Shr	
Gi0/2	Desg	BLK	4	128.3	Shr	
Gi0/3	Desg	BLK	4	128.4	Shr	
Gi1/0	Desg	FWD	4	128.5	Shr	Edge
Gi1/1	Desg	FWD	4	128.6	Shr	Edge

Коммутатор 5:

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 1

Address 0c68.b492.0000

Cost 4

Port 2 (GigabitEthernet0/1)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c9e.7547.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Altn	BLK	5	128.1	Shr	
-------	------	-----	---	-------	-----	--

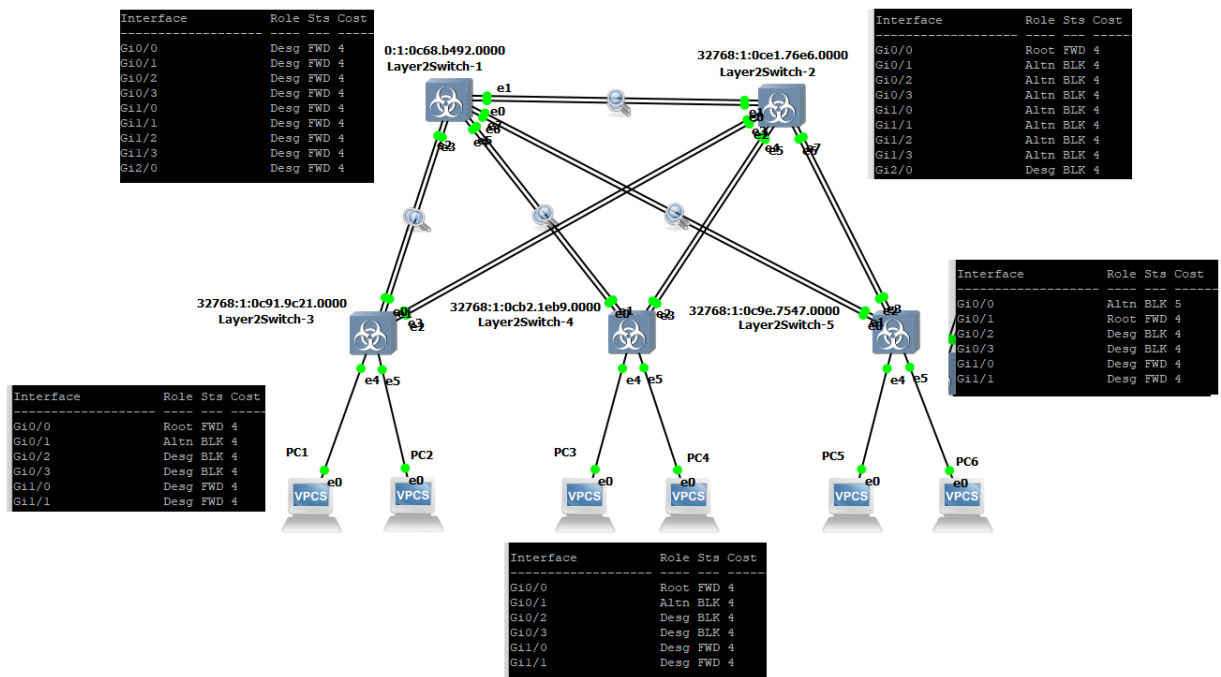
Gi0/1	Root	FWD	4	128.2	Shr	
-------	------	-----	---	-------	-----	--

Gi0/2	Desg	BLK	4	128.3	Shr	
-------	------	-----	---	-------	-----	--

Gi0/3	Desg	BLK	4	128.4	Shr	
-------	------	-----	---	-------	-----	--

Gi1/0	Desg	FWD	4	128.5	Shr	Edge
-------	------	-----	---	-------	-----	------

Gi1/1	Desg	FWD	4	128.6	Shr	Edge
-------	------	-----	---	-------	-----	------



Резервные порты отмечены как Altn.

7-4) При помощи wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет

К коммутатору 2:

e0:

No.	Time	Source	Destination	Protocol	Length	Info
4193	688.892733	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:91:9c:21:00:00 Cost = 4 Port = 0x8001
4194	689.480078	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 32768/100/0c:e1:76:e6:00:00 Cost = 0 Port = 0x8001[Malformed Packet]
4195	689.484417	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 32768/200/0c:e1:76:e6:00:00 Cost = 0 Port = 0x8001[Malformed Packet]
4196	689.485375	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 32768/300/0c:e1:76:e6:00:00 Cost = 0 Port = 0x8001[Malformed Packet]
4197	689.672236	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:91:9c:21:00:00 Cost = 4 Port = 0x8001
4198	689.891782	0c:68:b4:92:00:00	Nearest-Customer-Br...	STP	60	RST. TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8001
4199	690.476862	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 4 Port = 0x8001
4200	690.880661	0c:68:b4:92:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 32768/100/0c:68:b4:92:00:00 Cost = 0 Port = 0x8001[Malformed Packet]
4201	690.884410	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:b2:1e:b9:00:00 Cost = 4 Port = 0x8001
4202	690.885210	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:b2:1e:b9:00:00 Cost = 4 Port = 0x8001
4203	690.890756	0c:68:b4:92:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 32768/200/0c:68:b4:92:00:00 Cost = 0 Port = 0x8001[Malformed Packet]
4204	690.892324	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:9e:75:47:00:00 Cost = 4 Port = 0x8001
4205	690.892965	0c:68:b4:92:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 32768/300/0c:68:b4:92:00:00 Cost = 0 Port = 0x8001[Malformed Packet]
4206	691.480770	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 32768/100/0c:e1:76:e6:00:00 Cost = 0 Port = 0x8001[Malformed Packet]
4207	691.484765	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 32768/200/0c:e1:76:e6:00:00 Cost = 0 Port = 0x8001[Malformed Packet]
4208	691.486043	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 32768/300/0c:e1:76:e6:00:00 Cost = 0 Port = 0x8001[Malformed Packet]
4209	691.889076	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:e1:76:e6:00:00 Cost = 0 Port = 0x8001
4210	691.892483	0c:68:b4:92:00:00	Nearest-Customer-Br...	STP	60	RST. TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8001
4211	691.896137	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 4 Port = 0x8001
4212	692.477579	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 4 Port = 0x8001
4213	692.881366	0c:68:b4:92:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 32768/100/0c:68:b4:92:00:00 Cost = 0 Port = 0x8001[Malformed Packet]
4214	692.883819	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 8 Port = 0x8001
4215	692.886830	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:91:9c:21:00:00 Cost = 4 Port = 0x8001
4216	692.890361	0c:68:b4:92:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 32768/200/0c:68:b4:92:00:00 Cost = 0 Port = 0x8001[Malformed Packet]
4217	692.892690	0c:68:b4:92:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 32768/300/0c:68:b4:92:00:00 Cost = 0 Port = 0x8001[Malformed Packet]
4218	693.480439	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 32768/100/0c:e1:76:e6:00:00 Cost = 0 Port = 0x8001[Malformed Packet]
4219	693.484821	0c:e1:76:e6:00:00	Nearest-Customer-Br...	STP	60	RST. Root = 32768/200/0c:e1:76:e6:00:00 Cost = 0 Port = 0x8001[Malformed Packet]

e1:

e4:

e5:

No.	Time	Source	Destination	Protocol	Length	Info
24044	3631.365528	0c:68:b4:92:00:05	Nearest-Customer-Br...	STP	60 RST.	Root = 32768/100/0c:68:b4:92:00:00 Cost = 0 Port = 0x8006[Malformed Packet]
24045	3631.367429	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	TC + Root = 32768/1/0c:b2:1e:b9:00:00 Cost = 0 Port = 0x8002
24046	3631.369492	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	TC + Root = 32768/1/0c:9e:75:47:00:00 Cost = 8 Port = 0x8002
24047	3631.370561	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	TC + Root = 32768/1/0c:9e:75:47:00:00 Cost = 8 Port = 0x8002
24048	3631.376889	0c:68:b4:92:00:05	Nearest-Customer-Br...	STP	60 RST.	Root = 32768/200/0c:68:b4:92:00:00 Cost = 0 Port = 0x8006[Malformed Packet]
24049	3631.393490	0c:68:b4:92:00:05	Nearest-Customer-Br...	STP	60 RST.	Root = 32768/300/0c:68:b4:92:00:00 Cost = 0 Port = 0x8006[Malformed Packet]
24050	3631.398218	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	TC + Root = 32768/1/0c:9e:75:47:00:00 Cost = 8 Port = 0x8002
24051	3632.396403	0c:68:b4:92:00:05	Nearest-Customer-Br...	STP	60 RST.	TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8006
24052	3632.398666	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 4 Port = 0x8002
24053	3632.399322	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	Root = 32768/100/0c:b2:1e:b9:00:00 Cost = 0 Port = 0x8002[Malformed Packet]
24054	3632.401244	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	Root = 32768/200/0c:b2:1e:b9:00:00 Cost = 0 Port = 0x8002[Malformed Packet]
24055	3632.403069	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	Root = 32768/300/0c:b2:1e:b9:00:00 Cost = 0 Port = 0x8002[Malformed Packet]
24056	3633.365257	0c:68:b4:92:00:05	Nearest-Customer-Br...	STP	60 RST.	Root = 32768/100/0c:68:b4:92:00:00 Cost = 0 Port = 0x8006[Malformed Packet]
24057	3633.367376	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 12 Port = 0x8002
24058	3633.368200	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 12 Port = 0x8002
24059	3633.370175	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	TC + Root = 32768/1/0c:9e:75:47:00:00 Cost = 8 Port = 0x8002
24060	3633.376589	0c:68:b4:92:00:05	Nearest-Customer-Br...	STP	60 RST.	Root = 32768/200/0c:68:b4:92:00:00 Cost = 0 Port = 0x8006[Malformed Packet]
24061	3633.393222	0c:68:b4:92:00:05	Nearest-Customer-Br...	STP	60 RST.	Root = 32768/300/0c:68:b4:92:00:00 Cost = 0 Port = 0x8006[Malformed Packet]
24062	3633.397914	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	TC + Root = 32768/1/0c:9e:75:47:00:00 Cost = 8 Port = 0x8002
24063	3634.396087	0c:68:b4:92:00:05	Nearest-Customer-Br...	STP	60 RST.	TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8006
24064	3634.398342	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 4 Port = 0x8002
24065	3634.398999	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	Root = 32768/100/0c:b2:1e:b9:00:00 Cost = 0 Port = 0x8002[Malformed Packet]
24066	3634.401810	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	Root = 32768/200/0c:b2:1e:b9:00:00 Cost = 0 Port = 0x8002[Malformed Packet]
24067	3634.402752	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	Root = 32768/300/0c:b2:1e:b9:00:00 Cost = 0 Port = 0x8002[Malformed Packet]
24068	3635.364934	0c:68:b4:92:00:05	Nearest-Customer-Br...	STP	60 RST.	Root = 32768/100/0c:68:b4:92:00:00 Cost = 0 Port = 0x8006[Malformed Packet]
24069	3635.366819	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	TC + Root = 32768/1/0c:b2:1e:b9:00:00 Cost = 0 Port = 0x8002
24070	3635.368922	0c:b2:1e:b9:00:01	Nearest-Customer-Br...	STP	60 RST.	TC + Root = 32768/1/0c:91:9c:21:00:00 Cost = 8 Port = 0x8002

К коммутатору 5:

e6:

e7:

No.	Time	Source	Destination	Protocol	Length	Info
23798	3681.533518	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:91:9c:21:00:00 Cost = 8 Port = 0x8002
23799	3681.541424	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	RST. Root = 32768/300/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008[Malformed Packet]
23800	3682.172948	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002
23801	3682.192899	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 32768/100/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002[Malformed Packet]
23802	3682.193049	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 32768/200/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002[Malformed Packet]
23803	3682.193168	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 32768/300/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002[Malformed Packet]
23804	3682.194587	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002
23805	3682.544640	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	RST. TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
23806	3682.548044	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 4 Port = 0x8002
23807	3683.194389	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 4 Port = 0x8002
23808	3683.528102	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	RST. Root = 32768/100/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008[Malformed Packet]
23809	3683.529432	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	RST. Root = 32768/200/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008[Malformed Packet]
23810	3683.529688	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002
23811	3683.533632	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:91:9c:21:00:00 Cost = 8 Port = 0x8002
23812	3683.536705	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:91:9c:21:00:00 Cost = 16 Port = 0x8002
23813	3683.539680	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:91:9c:21:00:00 Cost = 24 Port = 0x8002
23814	3683.541663	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	RST. Root = 32768/300/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008[Malformed Packet]
23815	3683.543669	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:91:9c:21:00:00 Cost = 32 Port = 0x8002
23816	3684.192688	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 32768/100/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002[Malformed Packet]
23817	3684.192824	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 32768/200/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002[Malformed Packet]
23818	3684.192970	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. Root = 32768/300/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002[Malformed Packet]
23819	3684.194274	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. TC + Root = 32768/1/0c:9e:75:47:00:00 Cost = 0 Port = 0x8002
23820	3684.543929	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	RST. TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008
23821	3684.545459	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 4 Port = 0x8002
23822	3685.194110	0c:9e:75:47:00:01	Nearest-Customer-Br...	STP	60	RST. TC + Root = 0/1/0c:68:b4:92:00:00 Cost = 4 Port = 0x8002
23823	3685.528283	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	RST. Root = 32768/100/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008[Malformed Packet]
23824	3685.529282	0c:68:b4:92:00:07	Nearest-Customer-Br...	STP	60	RST. Root = 32768/200/0c:68:b4:92:00:00 Cost = 0 Port = 0x8008[Malformed Packet]

Все коммутаторы отправляют Hello-пакеты с корнем для VLAN1 другим коммутаторам, поэтому на линках с первым коммутатором есть входящие пакеты от других коммутаторов. Так же в RSTP быстрее реакция на потерю пакетов от корня и из-за этого коммутаторы чаще объявляют себя корнем иницируя перевыборы.

7-5) Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл

Коммутатор 5:

```
vIOS-L2-01#enable
```

```
vIOS-L2-01#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
vIOS-L2-01(config)#int gi 0/1
```

```
vIOS-L2-01(config-if)#spanning-tree vlan 1 cost 6
```

```
vIOS-L2-01(config-if)#exit
```

```
vIOS-L2-01(config)#exit
```

```
vIOS-L2-01#show spanning-tree vlan 1
```

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 1

Address 0c68.b492.0000

Cost 5

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

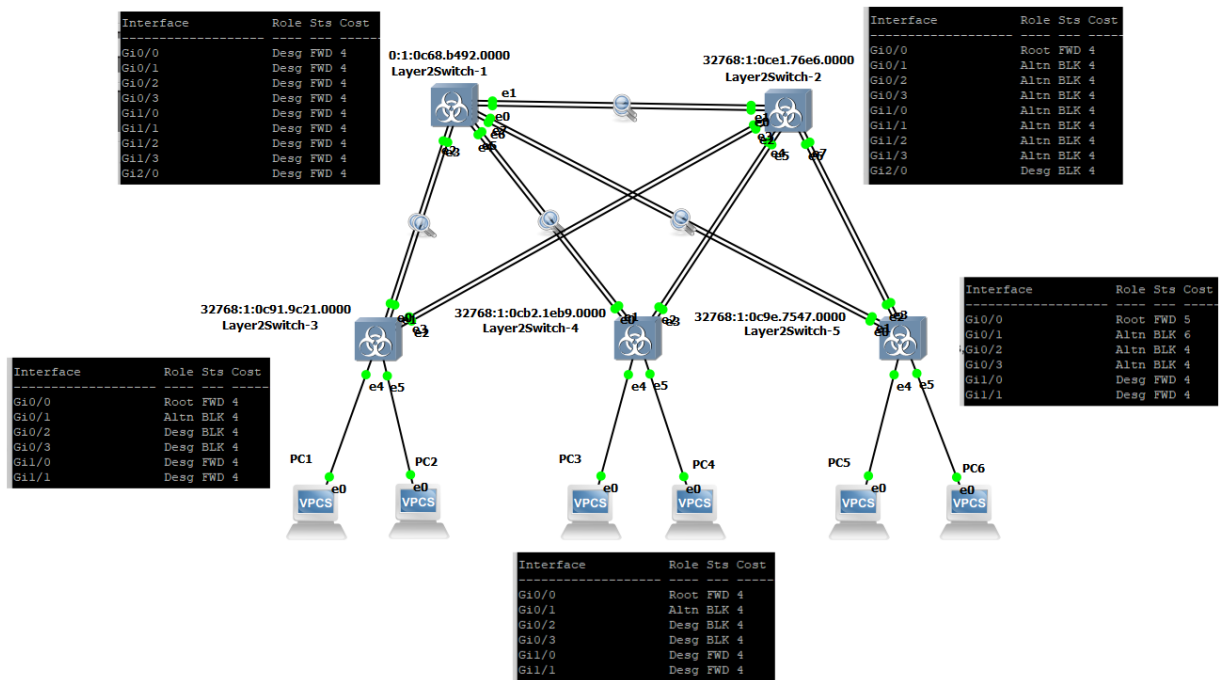
Address 0c9e.7547.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

AgingTime 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type

Gi0/0	Root	FWD	5	128.1	Shr	
Gi0/1	Altn	BLK	6	128.2	Shr	
Gi0/2	Altn	BLK	4	128.3	Shr	
Gi0/3	Altn	BLK	4	128.4	Shr	
Gi1/0	Desg	FWD	4	128.5	Shr	Edge
Gi1/1	Desg	FWD	4	128.6	Shr	Edge



Резервные порты отмечены как Altn.

7-6) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств